

What Drives the Exchange Rate?

Itskhoki & Mukhin (2023)

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Roadmap

- 1 Introduction
- 2 Model
- 3 Equilibrium
- 4 Results in the limit
- 5 Conclusions

Introduction - The Question

- "Exchange rate disconnect" $\implies$ challenging and persistent puzzles:
 - Meese & Rogoff puzzle $\implies$ ER moves like a random walk. »
 - PPP puzzle $\implies$ nominal and real ER go hand-to-hand. »
 - Backus-Smith puzzle $\implies$ corr. < 0 btw RER and Δ rel. consumption. »
 - Forward premium puzzle $\implies i_t - i_t^*$ predict appreciations. »
 - Financial disconnect $\implies$ low corr. btw prices of assets and ER. »
- **What types of shocks can or can't explain this puzzles?**
- We will answer this using a wedge-accounting framework.

Introduction - Foreshadowing the Results

- Standard macro-shocks $\implies$ no exchange rate disconnect.
- International shocks $\implies$ Backus-Smith and UIP still a puzzle.
- News about future macro shocks $\implies$ no financial disconnect.
- **All the "puzzles" are consistent with financial shocks.**
- This int. asset demand shocks are necessary and sufficient.

► Context

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Overview of the Model

- Two open economies model. Home (Europe) and Abroad (US).
- The nominal exchange rate $\mathcal{E}_t$ is the price of dollars in terms of euros. Higher $\mathcal{E}_t$ implies a nominal devaluation of the currency.
- Real exchange rate $Q_t = \mathcal{E}_t \frac{P_t^*}{P_t}$.
- We will focus on flexible exchange rates.
- A variety of shocks will hit the economy. In some cases those will be wedges that act as a proxy for unmodelled market imperfections.

$$\Omega_t = \{\pi_t, a_t, g_t, \chi_t, \kappa_t, \mu_t, \eta_t, \tilde{\zeta}_t, \psi_t\}$$

- Households maximize utility:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t e^{\chi_t} \left(\frac{1}{1-\sigma} C_t^{1-\sigma} - \frac{e^{\kappa_t}}{1 + \frac{1}{\varphi}} L_t^{1+\frac{1}{\varphi}} \right)$$

subject to their budget constraint:

$$P_t C_t + \sum_{j \in J_t} e^{\psi_t^j} \mathcal{P}_t^j B_{t+1}^j \leq \sum_{j \in J_{t-1}} \left(\mathcal{P}_t^j + \mathcal{D}_t^j \right) B_t^j + W_t L_t + \Pi_t + T_t$$

- Shocks χ_t and κ_t are inter- and intratemporal preference shifters.
- ψ_t^j are taxes or subsidies to the assets.

- HH supply labor according to static optimality condition:

$$e^{\kappa_t} C_t^\sigma L_t^{\frac{1}{\varphi}} = \frac{W_t}{P_t}$$

so κ_t can be interpreted as a labor wedge.

- They allocate C_t between home and foreign goods:

$$P_t C_t = P_{Ht} C_{Ht} + P_{Ft} C_{Ft}$$

- Assume CES between both (θ) and home bias (γ):

$$C_{Ht} = (1 - \gamma) e^{-\gamma \xi_t} \left(\frac{P_{Ht}}{P_t} \right)^{-\theta} C_t \text{ and } C_{Ft} = \gamma e^{-(1-\gamma) \xi_t} \left(\frac{P_{Ft}}{P_t} \right)^{-\theta} C_t$$

$$P_t = \left[(1 - \gamma) e^{-\gamma \xi_t} P_{Ht}^{1-\theta} + \gamma e^{(1-\gamma) \xi_t} P_{Ft}^{1-\theta} \right]^{\frac{1}{1-\theta}}$$

- HH choose asset positions optimally condition for every $j \in J_t$:

$$\beta \mathbb{E}_t \left\{ e^{\Delta \chi_{t+1} - \psi_t^j} \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \mathcal{R}_{t+1}^j \right\} = 1$$

- Where $\mathcal{R}_{t+1}^j$ is the next-period realized return on asset j

$$\mathcal{R}_{t+1}^j = \frac{\mathcal{P}_{t+1}^j + \mathcal{D}_{t+1}^j}{\mathcal{P}_t^j}$$

Model - Producers

- Production is realized with a linear technology:

$$Y_t = e^{a_t} L_t$$

- Production of domestic firms is distributed between home and foreign markets $Y_t = Y_{Ht} + Y_{Ht}^*$. Profits are distributed to the households:

$$\Pi_t = (P_{Ht} - e^{-a_t} W_t) Y_{Ht} + (P_{Ht}^* \mathcal{E}_t - e^{-a_t} W_t) Y_{Ht}^*$$

- Assume the following price-setting rules:

$$P_{Ht} = e^{\mu_t} e^{-a_t} W_t \text{ and } P_{Ht}^* = e^{\mu_t + \eta_t} e^{-a_t} \frac{W_t}{\mathcal{E}_t}$$

which are ad-hoc general pricing equations consistent with a broad range of price setting models.

- μ_t is a markup shock and η_t is a LOP shock (or trade cost shock).

Model - Government

- Government use lump-sum taxes to finance an exogenous stochastic path of government expenditure.

$$G_t = e^{g_t}$$

- Allocated between home and foreign goods in the same way as C_t .
- Government also collect taxes on financial positions, and returns net income to HH to have a balanced budget per period:

$$T_t = \sum_{j \in J_t} (e^{\psi_t^j} - 1) \mathcal{P}_t^j B_{t+1}^j - P_t G_t$$

- We assume monetary policy chooses the nominal price level, $P_t = e^{p_t}$ with $\Delta p_t = \pi_t$.

Table 1: Model parameters and shocks

Shocks / wedges		Parameters	
π_t, π_t^*	inflation (monetary) shock	$\beta = 0.99$	discount factor
a_t, a_t^*	productivity shock	$\sigma = 2$	relative risk aversion (inverse of IES)
g_t, g_t^*	government spending shock	$\varphi = 1$	Frisch elasticity of labor supply
χ_t, χ_t^*	intertemporal preference (deleveraging) shock	$\gamma = 0.15$	foreign share (home bias) parameter
κ_t, κ_t^*	labor wedge (sticky wages)	$\theta = 1.5$	elasticity of substitution
μ_t, μ_t^*	markup shock (sticky prices)	$\rho = 0.97$	persistence of shocks
η_t, η^*	law-of-one-price shock (LCP/DCP, trade costs)		
ξ_t, ξ_t^*	international good demand shock		
ψ_t^j, ψ_t^{j*}	financial (asset demand) shocks		

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Equilibrium

- Labor market clears in both countries.
- Goods market clear (analogous for foreign):

$$Y_{H_t} = C_{H_t} + G_{H_t} = (1 - \gamma) e^{-\gamma \zeta_t} \left(\frac{P_{H_t}}{P_t} \right)^{-\theta} [C_t + G_t]$$

$$Y_{H_t}^* = C_{H_t}^* + G_{H_t}^* = \gamma e^{(1-\gamma)\zeta_t^*} \left(\frac{P_{H_t}^*}{P_t^*} \right)^{-\theta} [C_t^* + G_t^*]$$

- Asset market clears:

$$B_{t+1}^j + B_{t+1}^{j*} = 0 \text{ for } j \in J_t \cap J_t^* \text{ and } B_{t+1}^j = B_{t+1}^{j*} = 0 \text{ if not}$$

- Country budget constraint:

$$\sum_{j \in J_t \cap J_t^*} \mathcal{P}_t^j B_{t+1}^j - \sum_{j \in J_{t-1} \cap J_{t-1}^*} \left(\mathcal{P}_t^j + \mathcal{D}_t^j \right) B_t^j = NX_t$$

$$NX_t = \mathcal{E}_t P_{H_t}^* Y_{H_t}^* - P_{F_t} Y_{F_t}$$

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Results in the limit $\gamma \rightarrow 0$

- Focus on the equilibrium system near the autarky limit ($\gamma \rightarrow 0$).
- $\gamma = 0$ would imply full trade autarky and thus a complete exchange rate disconnect. Exchange Rates have no impact on macro variables.
- Response of macro variables to exchange rates is increasing in γ .
- If there is no exchange rate disconnect near autarky, it will not be further away from autarky.
- $\gamma \approx 0$ is not a bad starting points for economies like the US, Japan or the Eurozone.
- For now, let's assume: $\psi_t^j - \psi_t^{j*} = \tilde{\psi}_t$ for all assets.

Macro Disconnect in the Limit

Definition 1: Macro Disconnect in the Limit

Denote $Z_t = (W_t, P_t, C_t, L_t, Y_t, R_t)$ a vector of all domestic macro variables, $\Omega_t = (\pi_t, a_t, g_t, \chi_t, \kappa_t, \mu_t, \eta_t, \tilde{\zeta}_t, \{\psi_t^j\})$ a vector of shocks and $\varepsilon_t = V'\Omega_t + V^{*'}\Omega_t^*$ an arbitrary combination of shocks. We say that an economy with $\gamma > 0$ exhibits macro disconnect in the autarky limit if:

$$\lim_{\gamma \rightarrow 0} \frac{dZ_t}{d\varepsilon_t} = 0 \text{ and } \lim_{\gamma \rightarrow 0} \frac{d\varepsilon_t}{d\varepsilon_t} \neq 0$$

Proposition 1

Proposition 1

The previous model cannot exhibit macro disconnect in the autarky limit if the combined shock ε_t in Definition 1 has zero weight on the subset of shocks $\{\eta_t, \eta_t^*, \xi_t, \xi_t^*, \tilde{\psi}_t\}$.

- Standard macro shocks $\Omega_t^M = (\pi_t, a_t, g_t, \chi_t, \kappa_t, \mu_t)$ together with Ω_t^{M*} , in any combination and with arbitrary cross-correlations, cannot reproduce an exchange rate disconnect property (even with $\gamma \rightarrow 0$).
- μ_t will affect wages, employment and output, κ_t will affect employment or consumption, a_t employment or output, g_t consumption or output and χ_t the risk-free rate.

► Sketch

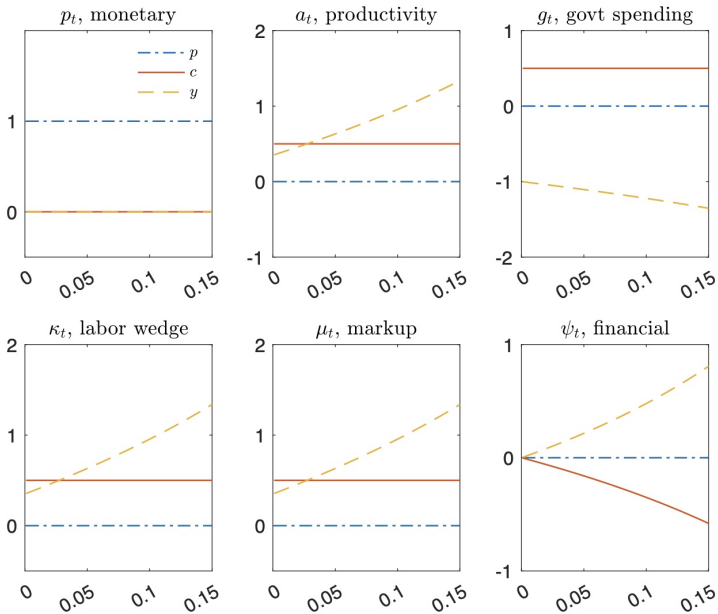


Figure 1: Relative macro and exchange rate volatility as a function of openness γ

Proposition 2

Proposition 2

Near the autarky limit, the international risk-sharing financial shock $\tilde{\psi}_t$ is the only shock in $\{\eta_t, \eta_t^*, \xi_t, \xi_t^*, \tilde{\psi}_t\}$ that simultaneously and robustly produces:

- ① positive correlation between terms of trade and real exchange rate;
 - ② negative correlation between real consumption growth and real exchange rate depreciation;
 - ③ deviations from UIP and a negative Fama coefficient.
- Both η_t and ξ_t will not be able to solve Backus-Smith and UIP puzzles. η_t will not have the right correlation between RER and ToT.

► Sketch

Definition 2

Denote with $F_t = \left(\{\mathcal{R}_t^i\}_{i \in \mathcal{A}_{t-1}}, \{\mathcal{R}_t^{i*}\}_{i \in \mathcal{A}_{t-1}^*} \right)$ a vector of asset returns that do not mechanically correlate with the exchange rate, and with $\varepsilon_t = V'\Omega_t + V^{*'}\Omega_t^*$ an arbitrary combination of shocks. We say that an open economy ($\gamma > 0$) exhibits financial disconnect in the autarky limit if

$$\lim_{\gamma \rightarrow 0} \frac{dF_t}{d\varepsilon_t} = 0 \text{ and } \lim_{\gamma \rightarrow 0} \frac{d\varepsilon_t}{d\varepsilon_t} \neq 0$$

Proposition 3

- Let's allow for specific shifters to risk-sharing per asset.
- Define two subset of assets, $\mathcal{A}_t$ and $\mathcal{A}_t^*$. They are assets whose returns do not depend mechanically on the exchange rate.
- $\mathcal{A}_t$ home-currency assets, its dividends is either a constant or an arbitrary function of local macro variables or local profits.

Proposition 3

Provided that the sets $\mathcal{A}_t$ and $\mathcal{A}_t^*$ are sufficiently rich, this model cannot exhibit financial disconnect ($\gamma \rightarrow 0$) if the combined shock ε_t has a weight of zero on the subset of shocks $(\eta_t, \eta_t^*, \xi_t, \xi_t^*, \{\psi_t^j, \psi_t^{j*}\}_{j \in J_t \cap J_t^*})$.

In particular, news shocks about the future macroeconomic fundamentals are inconsistent with the financial disconnect.

Intuition for Proposition 3

- Rewrite Euler equations of H and F as asset pricing equations:

$$\mathcal{P}_t^i = \mathbb{E}_t \sum_{\tau=1}^{\infty} \mathcal{M}_{t,t+\tau} \mathcal{D}_{t+\tau}^i e^{-\Psi_{t,t+\tau}^i}$$

$$\mathcal{P}_t^{j*} = \mathbb{E}_t \sum_{\tau=1}^{\infty} \mathcal{M}_{t,t+\tau}^* \mathcal{D}_{t+\tau}^{j*} e^{-\Psi_{t,t+\tau}^{j*}}$$

- $\mathcal{M}_{t,t+\tau}$ is the SDF and $\Psi_{t,t+\tau}^i$ the accum. asset demand shock.

$$\mathcal{M}_{t,t+\tau} = \beta^\tau e^{\chi_{t+\tau} - \chi_t} \left(\frac{C_{t+\tau}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}}$$

$$\Psi_{t,t+\tau}^i = \sum_{k=0}^{\tau-1} \psi_{t+k}^i$$

- Dividends and SDF depends on future macro shocks, they cannot generate financial disconnect. Int. shocks can generate it.

Proposition 4

Proposition 4

Home household demand shocks for foreign-currency assets, ψ_t^j for $j \in \mathcal{A}_t^*$, as well as foreign household demand for home-currency assets, ψ_t^{i*} for $i \in \mathcal{A}_t$, result in financial disconnect in the autarky limit.

- Take again previous equations. SDF and dividends of the local assets in autarky are determined solely by local shocks.
- Then, in autarky, prices of local assets $\mathcal{P}_t^i$ for $i \in \mathcal{A}_t$ too.
- The foreign Euler equation must hold:

$$\mathbb{E}_t \left\{ \mathcal{M}_{t,t+1}^* e^{-\psi_t^{i*}} \mathcal{R}_{t+1}^i \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right\} = 1$$

- Then ψ_t^{i*} must affect the exchange rate. Country-specific shocks won't get the disconnect.

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Conclusions

- The paper proposes a near-autarky limit as a convenient diagnostic tool to dissect shocks that drive the exchange rate.
- Traditional macroeconomic shocks, while important for business cycles, are inconsistent with the broad properties of macro exchange rate disconnect in the data.
- News shocks about future macro fundamentals are reflected in current asset prices, thereby violating the finance exchange rate disconnect.
- These findings underscore the critical role of financial shocks, particularly those related to shifts in foreign-investor demand for home-currency assets.

► Itskhoki & Mukhin (2021)

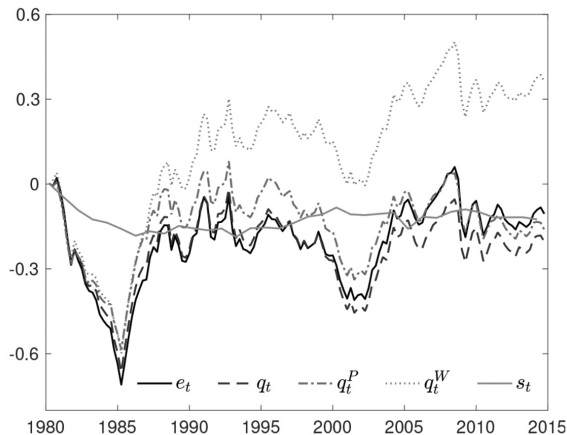
Meese & Rogoff Puzzle

- Nominal exchange rates follows a random-walk-like process, not robustly correlated with macroeconomic fundamentals.
- The exchange rate is an order of magnitude more volatile than macroeconomic aggregates such as consumption, output and inflation.
- All in all, short term nominal exchange rate variations are hard to explain even ex-post.
- No structural model posited by authors performs better in the out of sample prediction than a random walk.

► Back

PPP Puzzle

- There is a very high correlation between nominal and real exchange rates. Explanations on this phenomena based on nominal rigidities will indicate much lower persistence than observed in the data.



Backus-Smith Puzzle

- By combining the two Euler equations in intertemporal consumption:

$$\beta \mathbb{E}_t \left\{ e^{\Delta \chi_{t+1} - \psi_t^j} \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{P_t}{P_{t+1}} \mathcal{R}_{t+1}^j \right\} = 1$$

$$\beta \mathbb{E}_t \left\{ e^{\Delta \chi_{t+1}^* - \psi_t^{j*}} \left(\frac{C_{t+1}^*}{C_t^*} \right)^{-\sigma} \frac{P_t^*}{P_{t+1}^*} \mathcal{R}_{t+1}^j \frac{\mathcal{E}_t}{\mathcal{E}_{t+1}} \right\} = 1$$

- We get:

$$\frac{Q_{t+1}}{Q_t} = e^{\tilde{\psi}_t - \Delta \tilde{\chi}_{t+1}} \left(\frac{C_{t+1}/C_t}{C_{t+1}^*/C_t^*} \right)^\sigma$$

- Loosely speaking, USA and Canada should write contracts that imply transfers to Canada in states of nature where Canadian dollar is weak.
- This is not observed in the data.**

Forward Premium Puzzle

- Uncovered Interest Parity (UIP) is violated in the data. If it were true Fama's regressions should have a coefficient of 1. Its log-lin version:

$$\Delta e_t = i_t - i_t^*$$

- Cross-country interest rate differentials are not balanced out by expected depreciations but instead predicts exchange rate appreciations (with a low R^2).
- Hence currency carry-trades will have a positive expected return.

► Back

Financial Disconnect Puzzle

- Lack of exchange rate spanning in the asset markets.
- Using the full cross-section of asset returns (terms structure of bond returns and cross-section on equity returns) one can explain between 25% and 40% of contemporaneous exchange rate variation.
- Most exchange rate variations cannot be spanned by asset returns.

► Back

- Selected equations of the log-lin equilibrium:

$$w_t = \frac{p_t}{1-\gamma} - \frac{\gamma}{1-\gamma} p_{Ft}^* - \mu_t + a_t$$

$$\sigma c_t + \frac{1}{\varphi} l_t = w_t - p_t - \kappa_t$$

$$y_t - l_t = a_t$$

$$y_t - \varsigma c_t = 2\varsigma\gamma\tilde{c}_t + \frac{2\gamma(1-\gamma)\theta}{1-2\gamma} q_t + (1-\varsigma)[g_t - 2\gamma\tilde{g}_t]$$

$$r_{ft} = \mathbb{E}_t\{\sigma\Delta c_{t+1} + \Delta p_{t+1} - \Delta\chi_{t+1}\}$$

$$e_t = \sigma(\Delta c_t - \Delta c_t^*) + p_t - p_t^* + \zeta_t$$

$$\Delta\zeta_t = \tilde{\psi}_{t-1} - \Delta\tilde{\chi}_t$$

- Taking γ to 0 in the limit:

$$\lim_{\gamma \rightarrow 0} w_t = p_t - \mu_t + a_t = 0$$

$$\lim_{\gamma \rightarrow 0} \left\{ \sigma c_t + \frac{1}{\phi} l_t \right\} = a_t - \mu_t - \kappa_t = 0$$

$$\lim_{\gamma \rightarrow 0} \{y_t - l_t\} = a_t = 0$$

$$\lim_{\gamma \rightarrow 0} \{y_t - \varsigma c_t\} = (1 - \varsigma)g_t = 0$$

$$r_{ft} = \mathbb{E}_t \{ \sigma \Delta c_{t+1} + \Delta p_{t+1} - \Delta \chi_{t+1} \}$$

$$e_t = \sigma(c_t - c_t^*) + p_t - p_t^*$$

► Back

- Relationship between terms of trade and RER:

$$s_t = \frac{1}{1-2\gamma} q_t - \frac{2(1-\gamma)}{1-2\gamma} \tilde{\eta}_t$$

- An equation for $\tilde{c}_t$:

$$\begin{aligned} [(1-2\gamma)(\varphi\sigma + \varsigma) + 2\gamma\varphi\sigma] \tilde{c}_t &= (1+\varphi)\tilde{a}_t - \varphi\tilde{\mu}_t - \varphi\tilde{\kappa}_t \\ &- (1-2\gamma)(1-\varsigma)\tilde{g}_t + \gamma\varphi\tilde{\eta}_t + 2\gamma(1-\gamma)\tilde{\xi}_t - \frac{\gamma}{1-2\gamma} [2(1-\gamma)\theta + \varphi] q_t \end{aligned}$$

- From the linearized Euler equations for one-period risk-free bonds:

$$i_t - i_t^* = \mathbb{E}_t \{ \sigma(\Delta c_{t+1} - \Delta c_{t+1}^*) + (\Delta p_{t+1} - \Delta p_{t+1}^*) + \tilde{\psi}_t^f \} = \mathbb{E}_t \Delta e_{t+1}$$

- If all shocks but $\tilde{\psi}_t$ are zero:

$$\lim_{\gamma \rightarrow 0} \frac{\text{cov}(\Delta s_t, \Delta q_t)}{\text{var}(\Delta q_t)} = 1 > 0$$

$$\lim_{\gamma \rightarrow 0} \frac{1}{\gamma} \frac{\text{cov}(\Delta c_t - \Delta c_t^*, \Delta q_t)}{\text{var}(\Delta q_t)} = -\frac{2(2\theta + \varphi)}{\varphi\sigma + \varsigma} < 0$$

$$\lim_{\gamma \rightarrow 0} \gamma \frac{\text{cov}(\mathbb{E}_t \Delta e_{t+1}, i_t - i_t^*)}{\text{var}(i_t - i_t^*)} = -\frac{2\gamma\sigma(2\theta + \varphi)}{\varphi\sigma + \varsigma} < 0$$

- If $\tilde{\psi}_t = 0$:

- UIP will hold $\implies$ Fama puzzle still a puzzle.
- No negative correlation between RER and relative consumption.
- $\tilde{\eta}_t$ generates negative correlation between ToT and RER, $\tilde{\xi}_t$ positive.

- This work builds upon Itskhoki & Mukhin (JPE, 2021): "Exchange Rate Disconnect In General Equilibrium".
- Financial shocks plus significant home bias can explain all the puzzles.
- If you add standard macro shocks you can also explain business cycles.
- Nominal rigidities are not an essential part of the story.
- In this 2023 paper they don't perform a quantitative exercise since they already did that in this 2021 paper.

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Results: Proposition 1

- The equilibrium nominal exchange rate, $e_t = \log \mathcal{E}_t$, follows a volatile near-random-walk process.
- In particular, when both discount factor β and shock persistence $\rho \approx 1$, then $\text{corr}(\Delta e_t, \Delta e_{t+1}) \approx 0$.

► Back

Results: PPP Puzzle

- By recalling what we previously called the Price Block and the fact that monetary policy makes inflation zero in both economies:

$$q_t \equiv p_t^* + e_t - p_t = e_t$$

$$w_t - p_t = a_t - \frac{\gamma}{1 - 2\gamma} q_t$$

- By that we can characterize the producer-price and the wage-based real exchange rate as follows:

$$q_t^P \equiv p_{Ft}^* + e_t - p_{Ht} = \frac{1}{1 - 2\gamma} q_t$$

$$q_t^W \equiv w_t^* + e_t - w_t = \frac{1}{1 - 2\gamma} q_t - (a_t - a_t^*)$$

Results: PPP Puzzle

- If inflation is zero, this explains the comovement between e_t and q_t .
- The model will also produce a close comovement between consumer and producer relative prices.
- As γ decreases, q_t and q_t^P are not only tightly correlated but they will also have approximately the same volatility.
- ψ_t shock results in a near-perfect comovement between q_t and q_t^W .
- **Proposition 2:** With financial shocks, home bias, and $\beta, \rho \approx 1$, all real exchange rates exhibit a volatile near-random-walk behavior, with arbitrarily large half-lives, closely tracking the nominal exchange rate.

Results: Backus-Smith Puzzle

- We will explain this by emphasizing the role of the expenditure switching channel rather than the international risk-sharing.
- By using the quantity block, we get two equations:

$$\sigma c_t + \varphi y_t = (1 + \varphi) a_t - \frac{\gamma}{1 - 2\gamma} q_t$$

$$y_t = (1 - \gamma) c_t + \gamma c_t^* + 2\theta \frac{\gamma(1 - \gamma)}{1 - 2\gamma} q_t$$

- Then:

$$c_t - c_t^* = \kappa_a (a_t - a_t^*) - \gamma \kappa_q q_t$$

Results: Backus-Smith Puzzle

- **Proposition 3:** An equilibrium response to a financial shock ψ_t implies a negative comovement between the real exchange rate q_t and relative consumption $c_t - c_t^*$ with the relative volatility of the consumption response, $var(\Delta c_{t+1} - \Delta c_{t+1}^*) / var_t(\Delta q_{t+1})$ going to zero as $\gamma \rightarrow 0$.
- So ψ_t creates a negative correlation between real consumption and the real exchange rate, both in levels and in growth rates.
- If home bias present, the rel. vol. of the consumption response to ψ_t can be an order of magnitude smaller than the response of q_t .

► Back

Results: Forward Premium Puzzle

- From the intertemporal block we get:

$$i_t - i_t^* = \sigma \mathbb{E}_t \{ \Delta c_{t+1} - \Delta c_{t+1}^* \} = \sigma \kappa_a \mathbb{E}_t \Delta \tilde{a}_t - \gamma \sigma \kappa_q \mathbb{E}_t \Delta e_{t+1}$$

- Combining that with the UIP condition under the assumption $\chi_2 = 0$:

$$i_t - i_t^* = -\frac{\sigma \kappa_a}{1 + \gamma \sigma \kappa_q} (1 - \rho)(a_t - a_t^*) + \frac{\gamma \sigma \kappa_q}{1 + \gamma \sigma \kappa_q} \psi_t$$

$$\mathbb{E}_t \Delta e_{t+1} = -\frac{\sigma \kappa_a}{1 + \gamma \sigma \kappa_q} (1 - \rho)(a_t - a_t^*) - \frac{1}{1 + \gamma \sigma \kappa_q} \psi_t$$

► Back

Results: Forward Premium Puzzle

- Let's define the Fama regression as:

$$\mathbb{E}\{\Delta e_{t+1} | i_t - i_t^*\} = \hat{\beta}_F (i_t - i_t^*)$$

- Proposition 4:** $\hat{\beta}_{F|a} = 1$ but $\hat{\beta}_{F|\psi} = -\frac{1}{\gamma\sigma\kappa_q} \cdot \hat{\beta}_F$ as $\beta\rho \rightarrow 1$:
 - $\hat{\beta}_F \rightarrow -\frac{1}{\gamma\sigma\kappa_q} < 0$ and the R^2 in the Fama regression becomes arbitrarily small.
 - the volatility (persistence) of $(i_t - i_t^*)$ relative to Δe_{t+1} becomes arbitrarily small (large).
 - the Sharpe ratio of the carry-trade becomes arbitrarily small.

► Back

Results: Equilibrium Exchange Rate Dynamics

- **Proposition 5:** As $\beta\rho \rightarrow 1$:

- the exchange rate process becomes indistinguishable from a random walk, with $\mathbb{E}_t \Delta e_{t+1} \rightarrow 0$ and $\text{corr}(\Delta e_t, \Delta e_{t-1}) \rightarrow 0$.
- Volatility of exchange rates becomes unboundedly large relative to the volatility of the financial shock, $\frac{\text{var}(\Delta e_t)}{\text{var}(\psi_t)} \rightarrow +\infty$. Furthermore, under strong home bias $\gamma \approx 0$, the relative volatility of macroeconomic aggregates, such as GDP, is arbitrarily small.

► Back

Results: Equilibrium Exchange Rate Dynamics

- Fundamental shocks can also generate random-walk like behavior from exchange rates.
- But for ψ_t , as persistence grows, its effect on the exchange rate increases without bound, while for a_t its effect remains bounded.
- In response to a_t , GDP is more volatile than the exchange rate.
- a_t shocks generally cannot deliver exchange rate disconnect in volatilities, even in limiting economies.

► Back

Quantitative Analysis

- Add capital with adj. costs, int. inputs, nom. rigidities, variable mark-ups and conventional Taylor Rules. Previous results are robust.

» Calibration

- ψ shocks alone can fit well the exchange rate properties but not the real business cycle properties.
- Productivity and monetary shocks alone can explain real business cycle properties but not the exchange rates properties.
- Multishock models can explain both phenomena. Nominal rigidities do not change drastically the results (marginal better fit though).
- We can extend these models for asymmetries and small economies or different types of incomplete pass-through.

» Incomplete PT

» Small Economies

» Back

	DATA	SINGLE-TYPE SHOCKS				MULTISHOCK MODELS		
		Financial Shock		IRBC	NKOE	IRBC	IRBC ⁺	NKOE
MOMENTS	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Exchange rate disconnect:								
$\rho(\Delta e)$	≈ 0	-.02 (.09)	-.03 (.09)	.01 (.09)	-.06 (.09)	-.02 (.09)	-.03 (.09)	-.03 (.09)
$\sigma(\Delta e)/\sigma(\Delta gdp)$	5.2	17.7	9.8	.58	1.31	2.9	3.4	3.8
$\sigma(\Delta e)/\sigma(\Delta c)$	6.3	7.5	17.5	.96	2.18	3.9	5.8	6.6
B. Real exchange rate and the PPP:								
$\rho(q)$	.94	.93 (.04)	.91 (.05)	.94 (.04)	.82 (.06)	.93 (.04)	.91 (.05)	.90 (.04)
$\sigma(\Delta q)/\sigma(\Delta e)$	.99	.74	.98	1.40	.95	.83	.98	.97
$\text{corr}(\Delta q, \Delta e)$	.99	1.00	1.00	1.00	.99	.99	1.00	1.00
$\sigma(\Delta q^w)/\sigma(\Delta e)$	1.01	1.04	1.00	.56	.95	1.07	1.00	1.00
$\text{corr}(\Delta q^w, \Delta e)$	.99	1.00	1.00	-.99	.99	.96	1.00	1.00

MOMENTS	DATA (1)	SINGLE-TYPE SHOCKS				MULTISHOCK MODELS		
		Financial Shock		IRBC (4)	NKOE (5)	IRBC (6)	IRBC ⁺ (7)	NKOE (8)
		(2)	(3)					
C. Backus-Smith and forward premium:								
corr(Δq , $\Delta c - \Delta c^*$)	-.40	-1.00	-.95 (.01)	1.00	1.00	-.40 (.08)	-.40 (.08)	-.40 (.08)
Fama β	<0	-2.0 (1.4)	-3.4 (2.7)	1.6 (.8)	1.4 (.8)	-1.7 (1.5)	-2.2 (2.1)	-1.9 (1.3)
Fama R^2	.02	.03 (.02)	.03 (.02)	.08 (.03)	.03 (.02)	.02 (.02)	.02 (.02)	.02 (.02)
Carry trade SR	.20	.23 (.04)	.22 (.04)	.01 (.10)	-.03 (.09)	.23 (.04)	.22 (.04)	.22 (.04)
$\sigma(i - i^*)/\sigma(\Delta e)$	.06	.08	.05	.17	.12	.08	.06	.06
$\rho(i - i^*)$	.90	.93	.98	.95	.82	.94	.98	.94
$\rho(i)$	.97	.93	.98	.95	.83	.94	.98	.89
D. International business cycle moments:								
$\sigma(\Delta c)/\sigma(\Delta gdp)$	.82	2.37	.56	.60	.60	.74	.59	.57
corr(Δc , Δgdp)	.64	-1.00	-.82	1.00	1.00	.85	.80	.75
corr(Δz , Δgdp)	.81	-1.00	-.20	1.00	1.00	.81	.86	.86
corr(Δgdp , Δgdp^*)	.35	-1.00	-1.00	.35	.35	.35	.35	.35
corr(Δc , Δc^*)	.30	-1.00	-1.00	.40	.38	.14	.39	.40
corr(Δz , Δz^*)	.27	-1.00	-1.00	.45	.39	.10	.54	.55

TABLE 2
CONTRIBUTION OF ψ_t TO MACROECONOMIC VOLATILITY

	IRBC (1)	IRBC ⁺ (2)	NKOE (3)
Nominal exchange rate, $\text{var}(\Delta e)$ (%)	96	98	94
Real exchange rate, $\text{var}(\Delta q)$ (%)	87	97	94
Consumption, $\text{var}(\Delta c)$ (%)	20	10	12
GDP, $\text{var}(\Delta gdp)$ (%)	1	11	14

NOTE.—Variance decompositions (cols. 1–3) respectively correspond to the quantitative models in cols. 6–8 of table 1. The entries are the percent contributions of the financial shock ψ_t to the unconditional variances of the macro variables, with the remaining shares accounted for by the other shocks in each model specification.

Parameter	Variable	Value	Comment
Conventional parameters:			
Discount factor, quarterly	β	.99	
Relative risk aversion	σ	2	
Macro Frisch elasticity	$\nu = 1/\varphi$	1	
Intermediate share	ϕ	.5	
Capital share in value added	ϑ	.3	
Depreciation rate, quarterly	δ	.02	
CES parameter for labor (sticky wages)	ϵ	4	
Transmission mechanism:			
Trade openness	γ	.07	Trade-to-GDP ratio = .28
Elasticity of substitution	θ	1.5	Feenstra et al. 2018
Strategic complementarity	α	.4	Amiti et al. 2019
Monetary parameters:			
Taylor rule coefficient	ϕ_π	2.15	Clarida, Galí, and Gertler 2000
Interest rate smoothing	ρ_m	.95	$\rho(i_t), \rho(i_t - i_t^*) \approx .95$
Calvo probability for prices	λ_p	.75	Duration of four quarters
Calvo probability for wages	λ_w	.85	Duration of six quarters
Calibrated parameters:			
Persistence of shocks	ρ	.97	$\rho(gdp_t), \rho(i_t), \rho(\psi_t)$
Coefficient on NFA in UIP (16)	χ_2	.001	$\rho(\Delta b_t) \approx .95$
Standard deviation of the UIP shock	$\chi_1 \sigma_\psi$	1	Normalization
Standard deviation of productivity (monetary) shocks	$\sigma_a (\sigma_m)$	→	$\text{corr}(\Delta c_t - \Delta c_t^*, \Delta q_t) = -.4$
Correlation of productivity (monetary) shocks	$\varrho_{a,a^*} (\varrho_{m,m^*})$	→	$\text{corr}(\Delta gdp_t, \Delta gdp_t^*) = .35$
Capital adjustment cost parameter	κ	→	$\text{std}(\Delta z_t) / \text{std}(\Delta gdp_t) = 2.5$

Extensions with Incomplete Pass-Through

TABLE 3
TERMS OF TRADE AND NET EXPORTS

MOMENTS	DATA (1)	IRBC (2)	IRBC ⁺			NKOE		
			PCP (3)	LCP (4)	DCP (5)	PCP (6)	LCP (7)	DCP (8)
$\sigma(\Delta q) / \sigma(\Delta e)$	.99	.83	.85	.98	.91	.83	.97	.90
$\sigma(\Delta s) / \sigma(\Delta e)$	.25	.22	.86	.81	.08	.85	.81	.06
$\text{corr}(\Delta s, \Delta e)$	$\approx .20$	.98	.98	−.94	.59	.97	−.94	.63
$\sigma(\Delta nx) / \sigma(\Delta q)$	.10	.26	.27	.17	.22	.25	.16	.20
$\text{corr}(\Delta nx, \Delta q)$	≈ 0	.97	.99	.99	.99	.97	.97	.97

NOTE.—Additional moments and model specifications; IRBC and the two LCP columns correspond to cols. 6–8 in table 1.

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Extensions for small economies

TABLE 4
SMALL OPEN ECONOMY

	DATA						MODEL (IRBC ⁺)	
	United States (1)	Japan (2)	United Kingdom (3)	Korea (4)	Sweden (5)	New Zealand (6)	United States (7)	New Zealand (8)
MOMENTS								
Size (% world GDP),								
$\gamma^*/(\gamma + \gamma^*)$	23.7	10.5	4.1	1.3	.8	.2		
Imp/GDP (%), 2γ	12.1	11.5	24.4	33.4	33.0	22.8	12.1	22.8
Exp/GDP* (%), $2\gamma^*$	2.9	1.2	1.0	.5	.3	.1	2.9	.1
$\rho(q)$	.96	.96	.93	.92	.97	.96	.92	.91
$\sigma(\Delta q)/\sigma(\Delta e)$	.97	1.03	1.04	.95	.99	1.01	.99	.99
corr($\Delta q, \Delta c - \Delta c^*$)	-.22	.12	-.03	-.50	-.17	.01	-.40	-.40
$\sigma(\Delta e)/\sigma(\Delta gdp)$	4.5	4.5	4.9	3.3	2.9	2.1	5.3	3.6
$\sigma(\Delta nx)/\sigma(\Delta q)$	.06	.09	.16	.24	.25	.26	.15	.28

NOTE.—Empirical moments calculated using quarterly data for 1981–2017 from the World Bank's World Integrated Trade Solution database; exchange rates are trade-weighted bilateral exchange rates with major trading partners (see online app. A.1). Exp = export; Imp = import.